

COST-EFFECTIVE COASTAL VESSEL ALLOCATION

Presented by Team VisionX

TEAM MEMBERS



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PROBLEM STATEMENT:

Develop an optimal transportation model to minimize the total Transportation Cost of bulk cargo across the system.

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BACKGROUND:

This challenge focuses on reducing operational costs for HPCL's maritime logistics the coastal movement of bulk cargo.

1. **Goal:** Achieve cost-effective and efficient bulk cargo movement in HPCL's coastal trade operations.
2. **Fleet:** A fixed fleet of 9 coastal tankers is used for transportation.
3. **Network:** The operations involve 6 loading ports (unlimited supply) and 11 unloading ports (fixed demand).
4. **Cost Driver:** Total cost is calculated by multiplying the tanker's daily charter hire rate by the total number of days consumed for a complete trip.

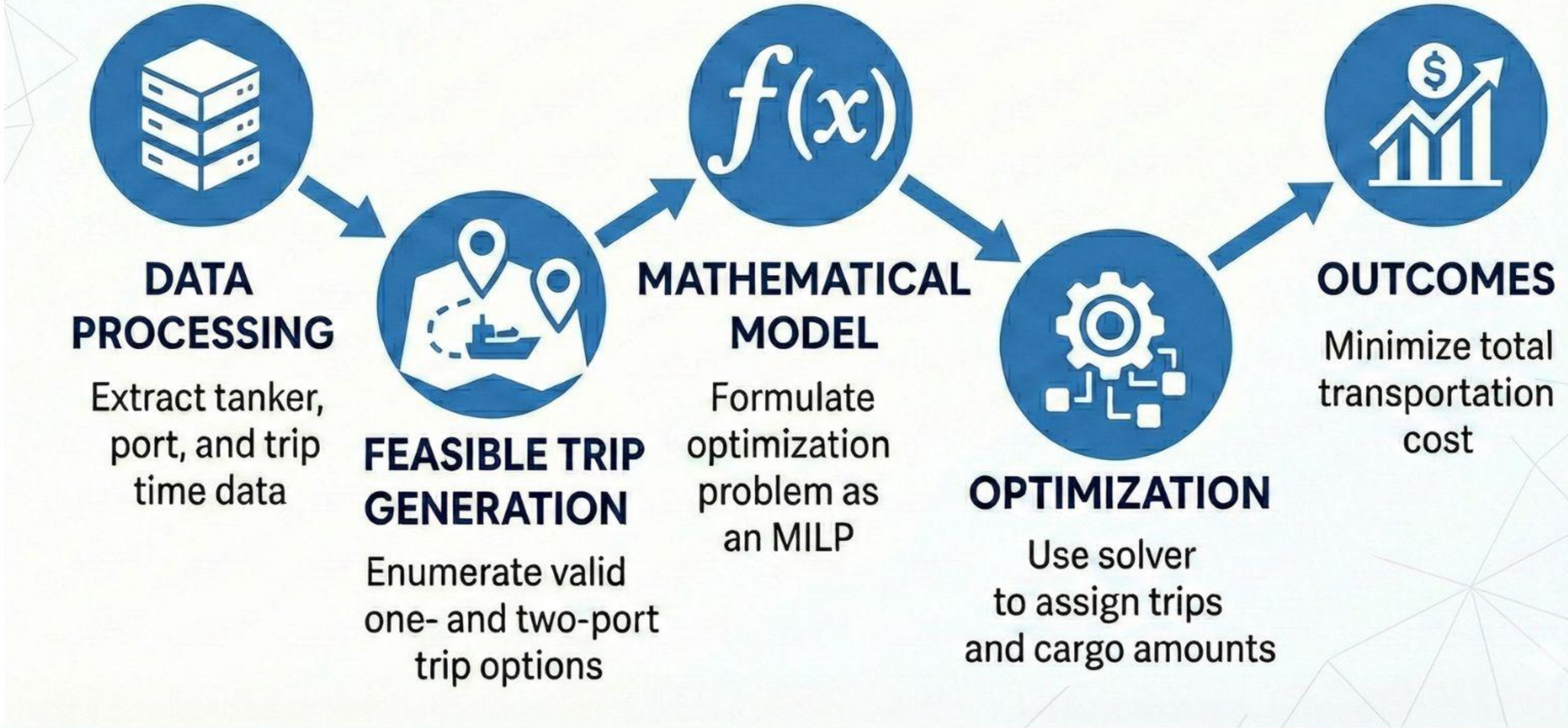
MANDATORY CONTRAINTS:

1. **Demand Satisfaction:** All demand at the 11 unloading ports must be met.
2. **Loading Rule:** Each tanker must take a full load from only one loading port per trip. (Single-port full loading).
3. **Discharge Rule:** Each complete trip may have a maximum of one or two discharge ports (multi-drop).
4. **Assignment:** The model must assign the optimal sequence of trips and cargo volume to each of the 9 tankers.

HOW OUR MODEL WORKS?

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SOLUTION ROADMAP



Stepwise Working Of the MILP Model For Optimizing Vessel Allocation:

1. DATA PROCESSING & ROUTE GENERATION



Inputs: Port Demands, Tanker Fleet, Distance Matrix.

Action: Calculates cost & duration for all possible trips:

- Single-Stop (L -> U)
- Multi-Drop (L -> U1 -> U2)

Calculates Total Trip Days & Cost for each valid combination.

2. MATHEMATICAL FORMULATION & CONSTRAINTS

OBJECTIVE FUNCTION

MINIMIZE TOTAL COST:
∑ (Tanker Rate × Trip Days)

DEMAND SATISFACTION:
∑ Cargo Delivered = Total Port Demand

CAPACITY LIMITS:
Total Cargo on Route ≤ Tanker Capacity

ROUTING RULES:
Max 2 Unloading Ports per Trip

3. OPTIMIZATION ENGINE (MILP SOLVER)

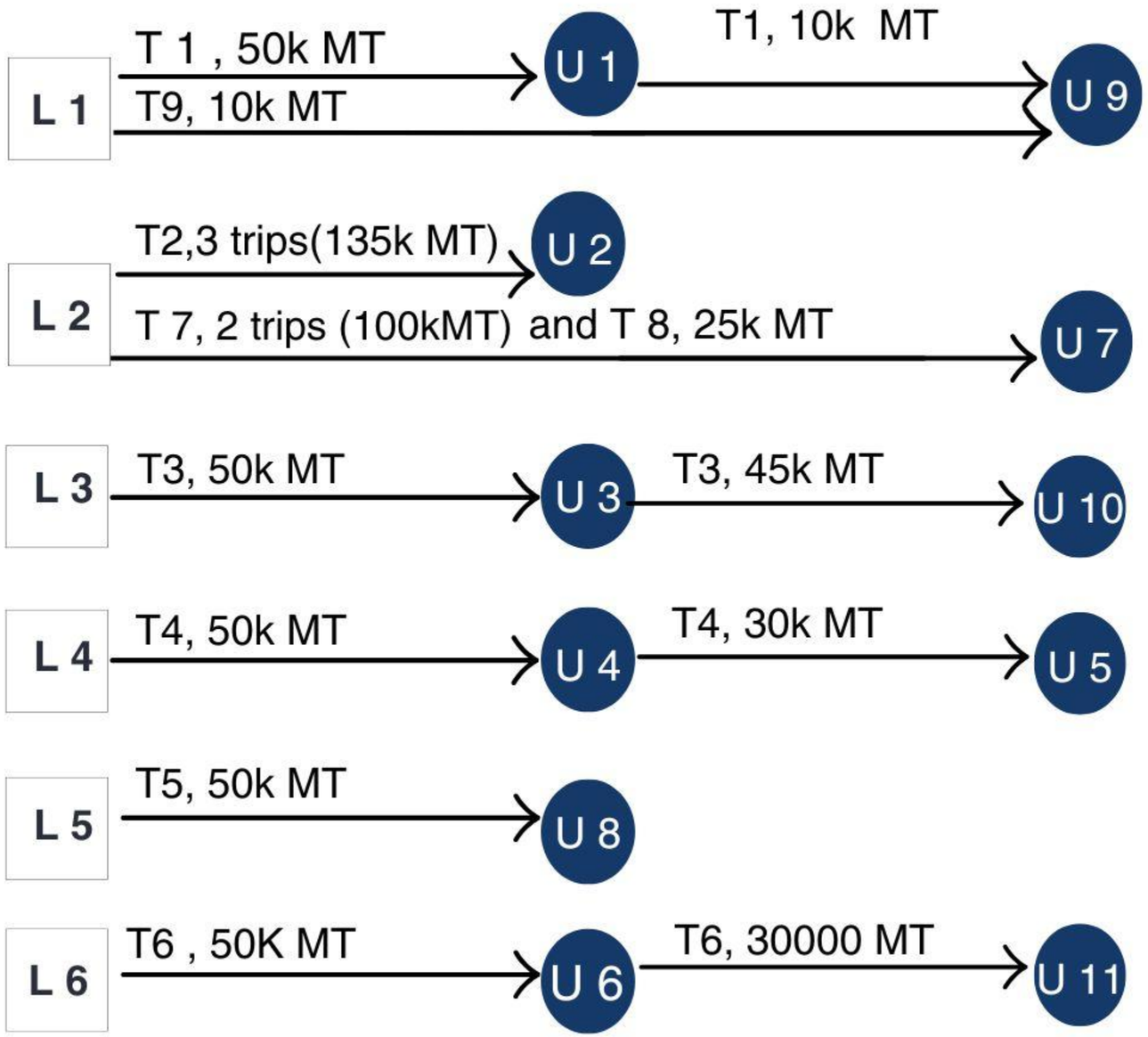


Action: The Mixed Integer Linear Programming solver evaluates thousands of trip combinations against all constraints.

OUTCOME: Finds the specific assignment of Tankers, Routes, and Cargo Splits that achieves the lowest overall cost, producing the optimal schedule.

1. Optimal Allocation of Tankers to Trips Using Model:

Specifying loading ports, discharge ports and trip sequence



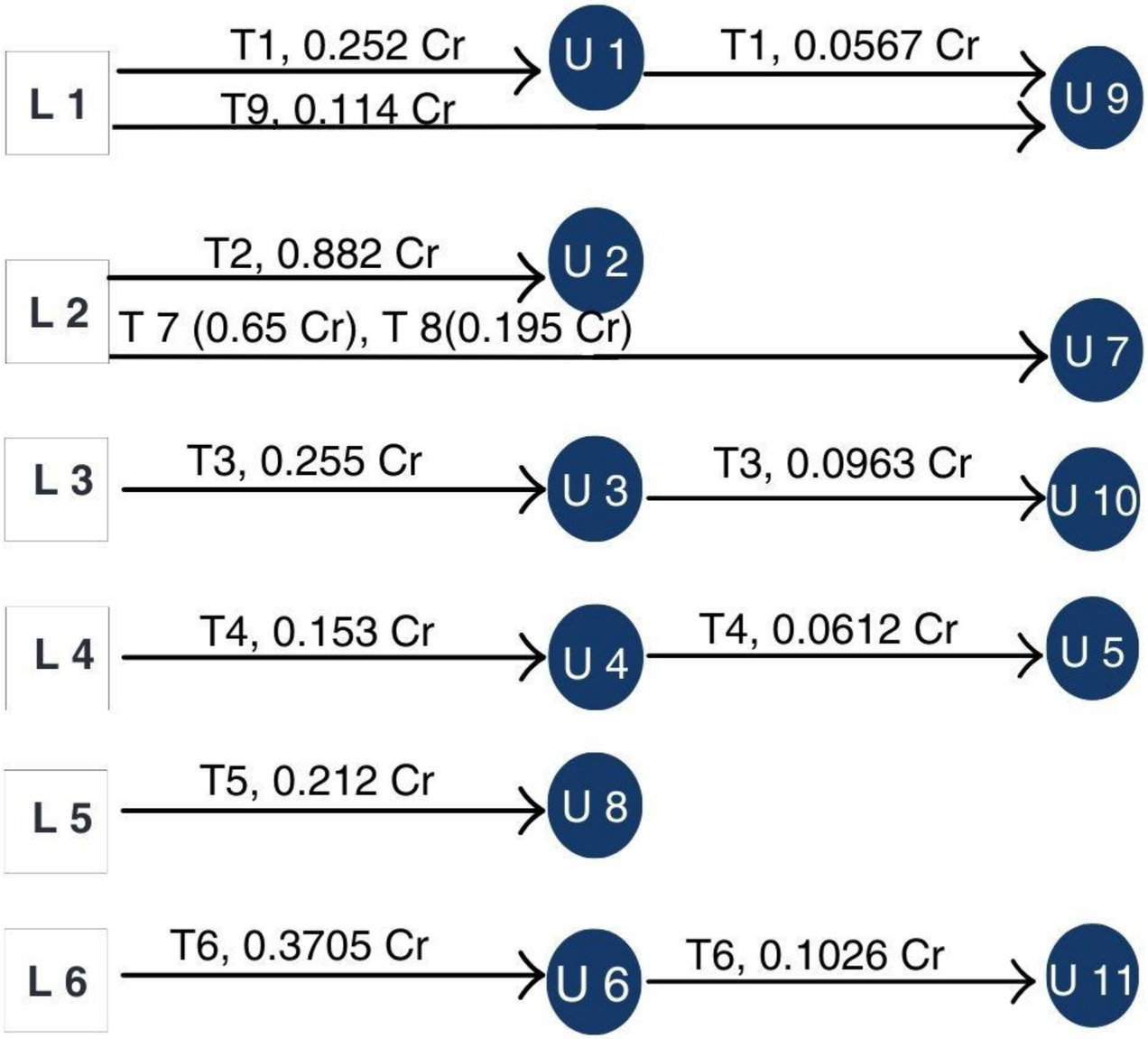
L - Loading Ports

U - Unloading Ports

T - Tanker

Demand Fulfillment	
Unloading Ports	Demand
U1	40k MT
U2	135k MT
U3	5k MT
U4	20k MT
U5	20k MT
U6	20k MT
U7	110k MT
U8	30k MT
U9	20k MT
U10	20k MT
U11	20k MT

2. Cost of Transportation of Fulfilling All Demands of All Unloading Ports:



L - Loading Ports

U - Unloading Ports

T - Tanker

Total Cost of Transportation is 3.4009 Cr

RESULT OF OUR MODEL

Tanker	Loading port	Route Sequence	Trips	Delivered(MT)	Trip Cost(Cr)
T1(50k)	L1	L1 → U1 → U9	1	50,000(U1 = 40000, U9 = 10000)	0.3087
T2(50k)	L2	L2 → U2	3	135,000	0.882
T3(50k)	L3	L3 → U3 → U10	1	25000 (U3 = 5000, U10 = 20000)	0.3519
T4(50k)	L4	L4 → U4 → U5	1	40,000(U4 = 20000 ,U5 = 20000)	0.2142
T5(50k)	L5	L5 → U8	1	30,000	0.212
T6(50k)	L6	L6 → U6 → U11	1	40000 (U6 = 20000, U11 = 20000)	0.4731
T7(50k)	L2	L2 → U7	2	100,000	0.65
T8(25k)	L2	L2 → U7	1	10,000	0.195
T9(25k)	L1	L1 → U9	1	10,000	0.114

Total Trip Costs(Cr) = 3.4009 Cr

Strategic Impact & Business Value

1. Maximized Cost Efficiency:

- **Minimum Possible Expenditure**: The model successfully identified the lowest cost baseline of 3.4009 Cr, ensuring no budget is wasted on inefficient routing.
- **Direct Savings**: Achieved a quantifiable decrease in total transportation costs by optimizing charter days and vessel selection.

2. Operational Excellence:

- **Precision Allocation:** Replaced manual scheduling with a mathematically optimal plan that assigns specific tankers (T1-T9) to exact routes.
- **Asset Optimization:** Ensured efficient utilization of the fixed fleet of 9 tankers, balancing high-capacity (50k) and lower-capacity (25k) vessels effectively .

3. Reliability & Future Growth:

- **100% Demand Satisfaction:** Guaranteed that every single metric ton of demand (455k Total) across all 11 unloading ports is met, eliminating shortage risks.
- **Scalable Architecture:** The MILP framework is designed to be a scalable solution, ready to adapt for future fleet expansions or new port additions.

4. Clearer Logistics Planning:

- Loading Ports: Where to pick up fuel (L1–L6).
- Discharge Rules: Precise instructions on whether to make a single stop or a multi-drop delivery.
- Trip Frequency: Exactly how many trips each tanker needs to make to meet the quota.

THANK YOU